

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Regular/Supplementary Winter Examination – 2024

Course: B.Tech

**Branch : Mechanical /Mechanical Engineering Sandwich/ Mechanical
And Automation Engg**

Semester : V

Subject Code & Name: Refrigeration & Air Conditioning (BTMPE504A)

Max Marks: 60

Date: 13/02/2025

Duration: 3 Hr.

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6.
4. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in () in front of the question.
5. Use of non-programmable scientific calculators is allowed.
6. Assume suitable data wherever necessary and mention it clearly.

		(Level/CO)	Marks
Q. 1	Objective type questions. (Compulsory Question)		12
1	The heat removing capacity of one tonne refrigerator is equal to a. 21 kJ/min b. 210 kJ/min c. 420 kJ/min d. 620 kJ/min	CO1 (L1)	1
2	The highest temperature during the cycle in a vapour compression refrigeration system occurs after a. compression b. condensation c. expansion d. evaporation	CO2 (L2)	1
3	How is the condensation process in vapour compression refrigeration cycle carried out? a. at constant volume b. at constant pressure c. at constant enthalpy d. all of the above	CO2 (L2)	1
4	Cooling effect is produced in the a. Condenser b. Expansion valve c. Evaporator d. None	CO2 (L2)	1
5	Due to sub-cooling in vapor compression refrigeration, COP a. Decreases b. Increases c. Remains same d. None	CO2 (L2)	1
6	Cascading refrigeration is applicable for a. Cold storage b. Comfort air conditioning c. Low temperature applications d. None	CO3 (L2)	1

7	An evaporator is also known as				CO2 (L2)	1
	a.freezing coil	b. cooling coil	c. chilling coil	d. all of these		
8	Due to intercooling in multistage refrigeration, COP				CO3 (L2)	1
	a.Increases	b. Decreases	c. Remains same	d. None		
9	The trade name of Refrigerant CH4 is				CO1 (L2)	1
	a. R150	b. R40	c. R50	d. R30		
10	COP of vapour Absorption system is _____ than vapour compression system				CO3 (L2)	1
	a.higher	b. greater	c.Lower	d. don't know		
11	Temperature recorded by ordinary thermometer is called as _____				CO4 (L1)	1
	a.DBT	b. WBT	c. DPT	d.None of the mentioned		
12	The amount of heat required to increase temperature by 1 degree is called as _____				CO4 (L1)	1
	a.latent heat	b.sensible heat	c.Specific heat	d. total heat		
Q. 2	Solve the following.					12
A)	Explain Simple air refrigeration system by using a line diagram.				CO1 (L2)	6
B)	Write trade names of following refrigerants i) CO2 ii) NH3 iii) C2H6 iv) CHCLF2 v) CCL2F2 vi) SO2				CO1 (L3)	6
Q.3	Solve the following.					12
A)	Derive the expression for C.O.P. of Vapour Compression cycle by using T-S & P-h diagram.				CO2 (L3)	6
B)	Illustrate by using P-h chart, any one method used to improve performance of simple VCC				CO2 (L3)	6
Q. 4	Solve Any Two of the following.					12
A)	Illustrate the three fluid vapour absorption system using a neat sketch.				CO3 (L2)	6
B)	By using P-h , explain the Cascade refrigeration system.				CO3 (L2)	6
C)	Illustrate the simple vapour absorption system using a neat sketch				CO3 (L2)	6

Q.5	Solve Any Two of the following.		12
A)	Define the terms i) Dry bulb temperature ii) Wet bulb temperature iii) Dew point temperature iv) Moist air v) dry air vi) saturated air	CO4 (L1)	6
B)	Write short notes on the split air conditioner.	CO4 (L2)	6
C)	On a particular day, surrounding condition are DBT =40 °C, WBT = 35 °C, Atmospheric Pressure 755 mm of Hg Calculate i) Dew point temperature ii) relative humidity iii) specific humidity iv) degree of saturation v) vapour density vi) enthalpy of mixture.	CO4 (L3)	6
Q. 6	Solve Any Two of the following.		12
A)	An air conditioning plant is required to supply 60 m ³ of air per minute at a DBT of 23°C and 50% RH. The outside air is at a DBT of 32°C and 60% RH. Determine the mass of water drained and capacity of cooling coil. Assume the air conditioning plant first to dehumidify and then to cool the air.	CO4 (L3)	6
B)	Describe various methods of noise control in AC	CO4 (L2)	6
C)	Illustrate Thermostatic expansion valve by using Neat sketch	CO4 (L2)	6
	*** End ***		